

TRAVIS AFB, CA, USA

WMO: 745160

Lat: 38.267N

Lon: 121.933W

Elev: 62

StdP: 14.66

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23202

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.3	33.2	17.3	13.2	53.8	21.2	16.0	52.1	29.0	52.2	25.9	51.8	3.8	30	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	30.0	99.2	67.5	95.4	66.6	91.4	65.6	70.0	93.6	68.4	90.9	67.0	88.4	10.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	81.7	72.3	60.1	77.8	71.5	58.9	74.6	70.9	33.8	93.7	32.5	91.1	31.4	88.1	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.8	26.8	25.0	DB	24.7	105.8	4.3	2.8	21.6	107.8	19.2	109.5	16.8	111.0	13.7	113.1
			WB	23.8	73.3	4.1	2.7	20.9	75.2	18.5	76.8	16.2	78.3	13.2	80.2

Monthly Climatic Design Conditions

			Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.9	47.7	50.9	55.3	58.6	64.1	70.1	72.8	73.1	71.5	64.4	54.3	47.7	(6)
(7)		DBStd	10.70	4.82	4.50	4.97	5.90	5.99	5.93	4.89	4.77	5.41	5.30	5.70	5.28	(7)
(8)		HDD50	284	101	39	11	3	0	0	0	0	0	0	23	107	(8)
(9)		HDD65	2483	535	394	302	205	88	14	1	1	8	75	324	536	(9)
(10)		CDD50	4273	31	66	176	262	438	602	707	715	644	445	151	36	(10)
(11)		CDD65	994	0	0	1	14	61	166	243	251	202	55	1	0	(11)
(12)		CDH74	12142	0	1	52	306	930	2191	2728	2710	2348	818	57	1	(12)
(13)		CDH80	5832	0	0	3	84	380	1099	1389	1395	1197	280	5	0	(13)
(14)	Wind	WSAvg	12.0	7.4	8.6	9.9	12.1	15.2	16.7	18.4	17.0	13.5	10.1	7.3	7.4	(14)
(15)	Precipitation	PrecAvg	21.7	4.1	4.0	2.8	1.2	0.8	0.2	0.0	0.0	0.1	1.1	2.0	4.3	(15)
(16)		PrecMax	35.4	13.3	12.9	10.0	4.2	3.1	2.0	0.4	0.6	0.6	5.0	6.3	12.0	(16)
(17)		PrecMin	11.2	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)
(18)		PrecStd	6.1	3.5	2.9	2.7	1.0	0.8	0.5	0.1	0.1	0.2	1.2	1.5	3.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.0	72.2	79.4	87.9	95.4	102.3	102.8	102.7	100.3	93.0	80.3	67.9	(19)
(20)		0.4%	MCWB	52.7	57.6	60.7	63.7	66.2	69.4	68.8	68.7	66.1	63.4	59.5	54.1	(20)
(21)		2%	DB	63.4	67.7	74.7	81.8	89.5	97.1	97.3	97.5	96.8	87.4	74.5	63.1	(21)
(22)		2%	MCWB	54.0	56.4	59.5	61.9	64.7	68.1	67.7	67.7	65.4	62.2	58.4	53.7	(22)
(23)		5%	DB	60.2	63.8	71.2	77.1	84.0	91.3	92.6	92.8	92.2	82.2	70.0	60.2	(23)
(24)		5%	MCWB	52.5	54.0	58.2	60.1	62.6	65.9	67.0	66.6	64.2	60.1	56.3	52.5	(24)
(25)		10%	DB	57.0	60.8	67.2	72.2	78.9	86.3	88.1	88.0	87.6	78.5	65.8	57.3	(25)
(26)		10%	MCWB	51.3	52.8	56.2	58.0	60.9	64.4	65.9	65.8	63.4	58.8	55.0	51.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.9	61.1	63.8	67.4	69.1	72.4	71.6	71.5	69.6	66.6	63.3	60.4	(27)
(28)		0.4%	MCDWB	62.3	66.7	74.1	82.8	91.0	96.9	98.0	97.1	92.5	84.5	71.1	62.9	(28)
(29)		2%	WB	56.3	58.8	61.4	64.1	66.3	69.3	69.2	69.2	67.5	64.4	61.1	57.5	(29)
(30)		2%	MCDWB	59.7	64.4	71.5	77.7	85.9	92.0	93.6	92.5	89.7	81.3	70.0	59.7	(30)
(31)		5%	WB	54.3	56.6	59.4	61.5	64.0	67.2	67.7	67.6	66.0	62.5	59.2	55.3	(31)
(32)		5%	MCDWB	57.5	61.5	68.3	74.2	81.5	89.3	90.3	89.5	87.3	76.8	66.4	57.5	(32)
(33)		10%	WB	52.4	54.6	57.4	59.3	61.8	65.1	66.2	66.1	64.5	61.0	57.2	53.3	(33)
(34)		10%	MCDWB	55.7	58.8	64.8	70.0	76.5	84.5	86.8	86.6	84.4	73.6	63.5	55.7	(34)
(35)	Mean Daily Temperature Range		MDBR	17.2	18.7	21.3	23.4	25.3	28.5	30.0	30.0	29.6	26.1	22.0	17.9	(35)
(36)		5% DB	MCDBR	23.8	24.9	28.3	31.8	34.1	36.9	36.6	37.4	37.6	33.8	29.3	22.0	(36)
(37)			MCWBR	15.2	14.9	15.9	15.6	13.8	13.4	12.7	12.7	12.7	14.1	15.2	13.8	(37)
(38)			MCDBR	16.6	20.3	24.6	27.9	31.3	34.2	33.8	33.4	32.3	28.1	23.2	13.9	(38)
(39)		5% WB	MCWBR	12.5	13.6	14.6	14.9	13.9	13.5	12.8	12.5	12.0	12.5	13.7	10.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.322	0.335	0.355	0.360	0.359	0.360	0.346	0.345	0.329	0.323	0.326	(40)	
(41)		taud	2.526	2.524	2.473	2.398	2.408	2.417	2.407	2.437	2.445	2.520	2.518	2.511	(41)	
(42)		Ebn at Noon	274	289	294	293	292	291	289	292	286	282	270	262	(42)	
(43)		Edh at Noon	24	28	32	37	37	37	37	35	33	27	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	715	1034	1469	1938	2320	2575	2534	2264	1859	1325	871	634	(44)	
(45)		RadStd	121	132	147	163	138	105	72	71	63	87	60	102	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	-1.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	+0.68	N/A	-2.02	N/A	N/A	N/A	-211	+236	N/A

Nomenclature: See separate page